

Noor Khial

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SUMMARY

Applied AI/ML engineer and researcher with experience in designing and deploying production machine learning systems for **logistics and customs operations**. Holds an MSc in Computing from Qatar University, with research on online learning algorithms for autonomous aerial navigation.

Fun fact: one reviewer left 99 comments on a single submission of mine. Since the journal was among the most cited in AI, I responded with a 60+ page rebuttal and the paper was accepted. Persistence pays off.

INDUSTRY EXPERIENCE

- **AI/ML Engineer** May 2025 – Present
Malomatia — Authority of Customs, Qatar (Al-Nadeeb System)
 - **HSSify Engine — HS Code Classification System** (*1st place, Malomatia AI Contest*). in-house AI system that predicts the Harmonized System (HS) code of imported goods. Built on *Microsoft Azure* with *fine-tuned BERT-family transformer models* via Hugging Face. Currently in production within the Al-Nadeeb customs platform, where it achieves **92% accuracy** and an **80% efficiency improvement** over manual classification, measured on live production traffic.
 - **AI-Based Goods Valuation System** (*in progress*). end-to-end ML pipeline that predicts the price of imported items. The pipeline operates in three stages:
 1. *Data acquisition*: automated data collection pipeline to extract raw goods data and assemble it into structured object.
 2. *Model training*: trained meta deep learning models on the structured data to capture cross-category pricing patterns.
 3. *Price prediction*: integrated transformer-based architectures into the predictive head of the meta-learner.Current production performance: **71% accuracy** with a **60% improvement** over the manually configured baseline.
 - **Stack**: Microsoft Azure (Document Intelligence, Computer Vision, Vector Search, Functions, Web Services), PyTorch, Hugging Face Transformers, BERT, Docker, SQL, Python.
- **AI Engineer** Feb 2025 – Apr 2025
Applab
 - Built multi-agent AI chatbots for Qatari national institutes on Microsoft Azure, integrating Azure OpenAI Service with retrieval and orchestration components.
- **Full Stack Intern** May 2022 – Jul 2022
QCRI, Cybersecurity Division (CIBR)
 - Developed a responsive web application for personalized investment portfolio generation using ReactJS and RESTful APIs.

EDUCATION

- **MSc in Computing** Jan 2023 – Dec 2024
Qatar University GPA: 3.77/4.0
 - *Thesis*: Smart Path Planning for Mobile Multi-Target Visitation in UAV Missions in Stochastic & Adversarial Environments
 - *Supervisor*: [Prof. Amr Mohamed](#), Professor of Computer Engineering and Head of the CSE Department at Qatar University.
 - *Funding*: Graduate Research Assistantship (tuition + stipend)
 - *Award*: Research Distinguished Award
 - *Coursework*: Algorithms, Deep Learning, Information Retrieval, Computer Vision
- **BSc in Computer Engineering** Jan 2018 – May 2022
Qatar University GPA: 3.5/4.0
 - *Senior Project*: Design and Implementation of an Integrated Onboard Computer & Transceiver Subsystem for [QUBESat](#), the first CubeSat launched by Qatar under Qatar University.
 - *Supervisor*: [Prof. Tamer Khattab](#), Professor of Electrical Engineering, Head of the Intelligent Information Processing Lab (IIPL), and the Excellence in Teaching Department at Qatar University.
 - *Funding*: The Amiri Diwan Scholarship (merit-based, full tuition)
 - *Coursework*: Probability & Statistics, Discrete Mathematics, Machine Learning

PUBLICATIONS

Peer-Reviewed Journal Papers

- [J1] **Khial, N.**, Allahham, M.S., Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., “DRONE-RL: Dynamic reinforcement learning for online navigation of UAVs in evolving environments,” *Knowledge-Based Systems*, Elsevier, vol. 334, p. 115147, 2026.
- [J2] **Khial, N.**, Mhaisen, N., Mabrok, M., and Mohamed, A., “An online learning framework for UAV search mission in adversarial environments,” *Expert Systems with Applications*, Elsevier, vol. 267, p. 126136, 2025.

Conference Papers

- [C1] **Khial, N.**, Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., “Multi-Target Path Planning with Probabilistic Detection in Cluttered Environments,” *IEEE International Conference on Communications (ICC)*, pp. 1298–1303, 2025.

RESEARCH EXPERIENCE

- **Part-Time Researcher** Sep 2025 – Present
[Qatar University](#)
 - Applying new online learning theory to autonomous navigation; manuscripts under review.
 - Concurrent with full-time AI/ML engineering role at Malomatia.
- **Graduate Research Assistant** Jan 2023 – Jan 2025
[Qatar University](#)
 - Contributed to multiple funded research projects.
 - Designed online learning and deep RL algorithms for UAV navigation in non-stationary settings.
 - Published 2 journal papers in top ranked AI journals.
- **NLP Research Intern** May 2021 – Jul 2021
[QCRI, Arabic Language Technologies](#)
 - Fine-tuned AraBERT for Arabic propaganda detection, achieving **85% accuracy**.
 - Built an automated data collection and preprocessing pipeline for Arabic text.
- **Research Intern** Jun 2020 – Jul 2020
[Center for Advanced Materials \(CAM\)](#)

OPEN-SOURCE RESEARCH SOFTWARE

- **Arabic Information Retrieval System** [GitHub](#)
Semantic search system combining the FAISS engine with the mContriever deep model for Arabic document retrieval.
Python, FAISS, PyTorch, Hugging Face Transformers, OpenAI API
- **UAV Navigation in Adversarial Environments** [GitHub](#)
Deep reinforcement learning implementation with a custom simulation environment for multi-target tracking and path optimization.
Python, TensorFlow, OpenAI Gym

ACADEMIC SERVICE

- **Technical Program Committee Member:** IEEE WCNC, IEEE ENERGYCON, IEEE ICC
- **Conference Volunteer:** ASTROCON-2021, IEEE-ICIOT-2020

TEACHING EXPERIENCE

- **Academic Tutor** Sep 2020 – May 2021
[Qatar University](#)
 - Tutored undergraduate students in Calculus II & III and Probability & Statistics.

AWARDS & HONORS

- **1st Place, Malomatia AI Contest** – HSSify Engine, 2025
- **Graduate Research Assistantship** – Qatar University, 2023–2024
- **The Amiri Diwan Scholarship** – 2018–2022
- **Dean’s List** – Qatar University, College of Engineering, 2019–2022
- **QU Undergraduate Startups Pre-Incubation Program** – Qatar University, 2022

TECHNICAL PROFICIENCIES

Research Areas	Online Learning	Deep RL	Meta-Learning	NLP	Autonomous Navigation	Applied ML
ML/DL Frameworks	PyTorch	TensorFlow	Hugging Face Transformers	BERT family	FAISS	OpenAI Gym
Cloud & MLOps	Microsoft Azure	Document Intelligence	Computer Vision	Vector Search	Azure Functions	
	Azure Web Services	Azure OpenAI	Docker			
Languages	Python	SQL	C	JavaScript	MATLAB	

LANGUAGES

Arabic – Native English – Fluent